Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

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Derivatives: Options

(Chapter 24)

Outline of today

Options

1. Definition

2. Types of options

3. Payoff and profit

4. Prices and determinants

5. Options Vs Futures

Options: Definition

- ✓ An **option** gives the holder the **right** to buy (or sell) an *underlying asset in the future at a pre-determined price* (the ***strike*** or ***exercise*** *price*), *at a pre-determined date* (at ***expiration***)

- ✓ Unlike a forward/future contract …
  - … the **holder** (or **purchaser**) of an **option** **is not obliged** to make the purchase/sale of the underlying asset if this is unfavorable to him

- ✓ Like a forward/future contract…
  - the **writer** **(**or **seller**) of an **option** **has** the **obligation** of complying with these terms and cannot refuse to buy/sell the asset (under unfavorable conditions)

Options: Definition

- ✓ In other words, there is a fundamental **asymmetry** between the rights of ***holder*** of an option vs. those of a ***writer***

- ✓ To compensate for his/her weaker rights, the seller/writer of an option will demand a compensation, called the **option premium** (effectively, the *price* of the option)

- ✓ The buyer of the option will always have to pay the premium, no matter if she exercises the option or not

Types of Options

There are two main types:

1. **Call option**: gives the *holder* the right to *buy* the underlying asset at the strike price within a specified period of time

2. **Put option**: gives the *holder* the right to *sell* the underlying asset at the strike price within a specified period of time

Note that you can buy/sell either type! So, it gets complicated

	Holder	***Writer***
	Long (buy)	**Short (sell)**
CALL	Right to buy	Obligation to sell
PUT	Right to sell	Obligation to buy

Options: Simple Example 1

Before we get into more details, let's work out a simple example.

✓ Suppose you <u>have</u> a call option with a strike price of $100, maturing tomorrow (analogous of delivery date in forwards) and a premium of $5.

1. The stock tomorrow is worth $120

    Q: Do you exercise the option? What is your profit?

Options: Simple Example 1

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✓ Suppose you have a call option with a strike price of $100, maturing tomorrow (analogous of delivery date in forwards) and a premium of $5.

1. The stock tomorrow is worth $120

    Q: Do you exercise the option? What is your profit?

    A: Yes, you exercise it because it is profitable:

    You can buy at $100 and sell spot at $120

    Gain $120- $100-$5=$15

Options: Simple Example 1

Before we get into more details, let's work out a simple example.

✓ Suppose you have a call option with a strike price of \$100, maturing tomorrow (analogous of delivery date in forwards) and a premium of \$5.

2. The stock tomorrow is worth \$90

Q: Do you exercise the option? What is your profit?

Options: Simple Example 1

Before we get into more details, let's work out a simple example.

✓ Suppose you <u>have</u> a call option with a strike price of $100, maturing tomorrow (analogous of delivery date in forwards) and a premium of $5.

2. The stock tomorrow is worth $90

Q: Do you exercise the option? What is your profit?

A: No, you do not

You could buy at $100 but then sell at $90:

avoid loss of $10 but still need to pay the premium: loss of $5

Options: Simple Example 2

Another simple example of a holder of an option

- ✓ Suppose you have a put option with a strike price of $250, maturing tomorrow and a premium of $10.

1. The stock tomorrow is worth $210

   Q: Do you exercise the option? What is your profit?

Options: Simple Example 2

Another simple example of a holder of an option

✓ Suppose you <u>have</u> a put option with a strike price of \$250, maturing tomorrow and a premium of \$10.

1. The stock tomorrow is worth \$210

   Q: Do you exercise the option? What is your profit?

   A: Yes, you exercise it because it is profitable:

   You can buy spot (if you don’t have it yet!) at \$210 and deliver at \$250

   Gain \$250 - \$210-\$10=\$30

Options: Simple Example 2

Another simple example of a holder of an option

✓ Suppose you have a put option with a strike price of $250, maturing tomorrow and a premium of $10.

2. The stock tomorrow is worth $325

Q: Do you exercise the option? What is your profit?

Options: Simple Example 2

Another simple example of a holder of an option

✓ Suppose you have a put option with a strike price of $250, maturing tomorrow and a premium of $10.

2. The stock tomorrow is worth $325

Q: Do you exercise the option? What is your profit?

A: No, you do not

You could buy at $325 but then sell at $250:

avoid loss of $75 but still need to pay the premium: loss of $10

Options Payoff vs Profit

- ✓ The option (gross) **payoff** is the amount the option pays at expiration
- ✓ We call the option payoff the *option* ***value***
- ✓ The option **profit** is the (gross) payoff …
  - ○ … **minus** the premium paid (*long* side: holder/purchaser)

    or
  - ○ … **plus** the premium received (*short* side: writer/seller)

Profit for *holder* (long)

- ✓ **Call option**: exercise a call option only if buying at the strike price *X* is more favorable than spot price *S* (S>X)
  - Payoff is the difference between the value *X* and *S*, and in any case never below zero (because if *S* < *X* , you don't exercise the option)

$$\pi_{long}^{call} = \underbrace{\max(S - X, 0)}_{\text{Call Payoff}} - p$$

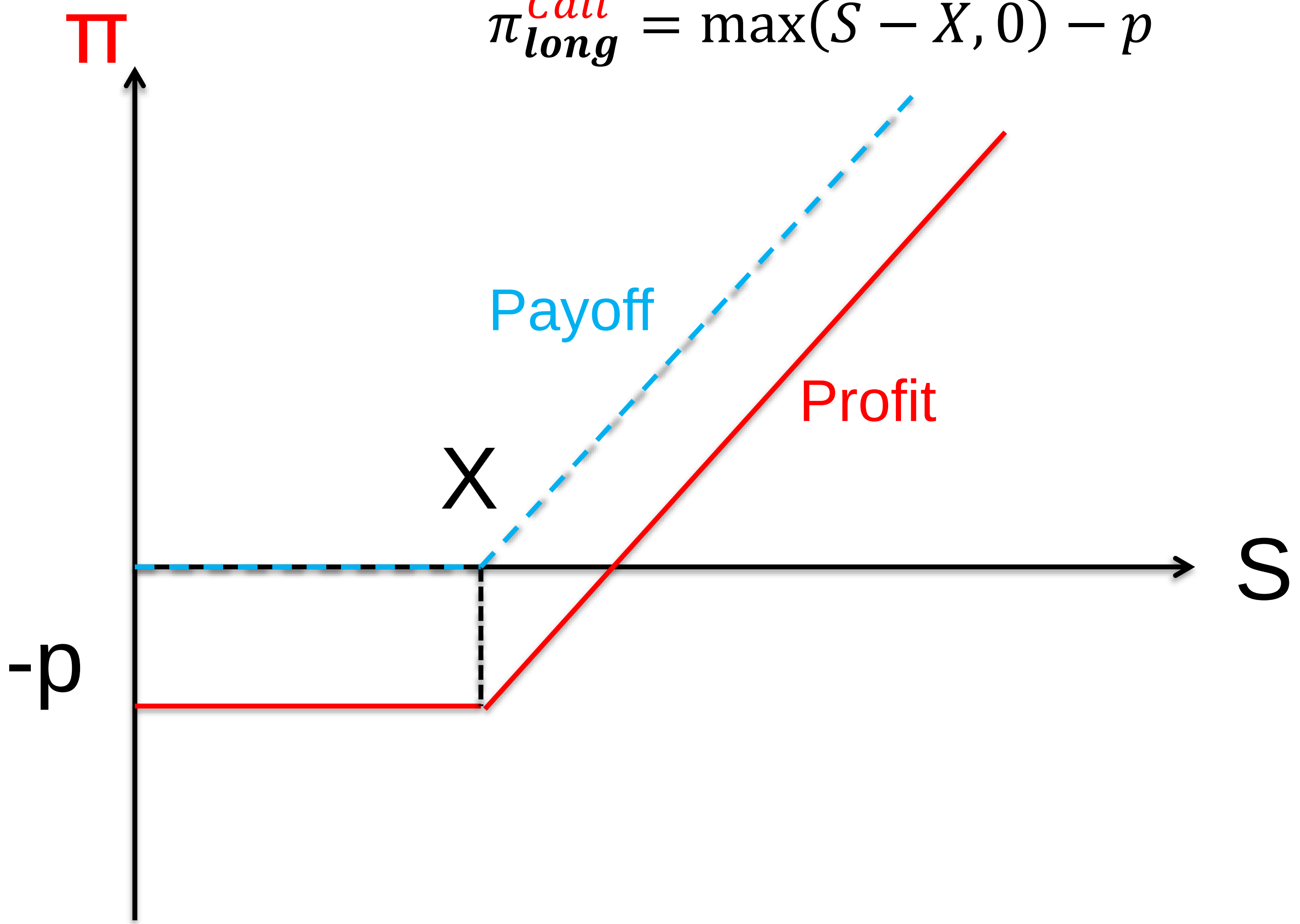
To get the profit subtract p

- ✓ **Put option**: exercise a put option only if selling at the strike price *X* is more favorable than the spot price *S* (X>S)
  - again, payoff (=*X-S*) is never below zero (because if *S*>*X* , you don't exercise the option)

$$\pi_{long}^{put} = \underbrace{\max(X - S, 0)}_{\text{Put Payoff}} - p$$

Profit from *holding* (long) a Call

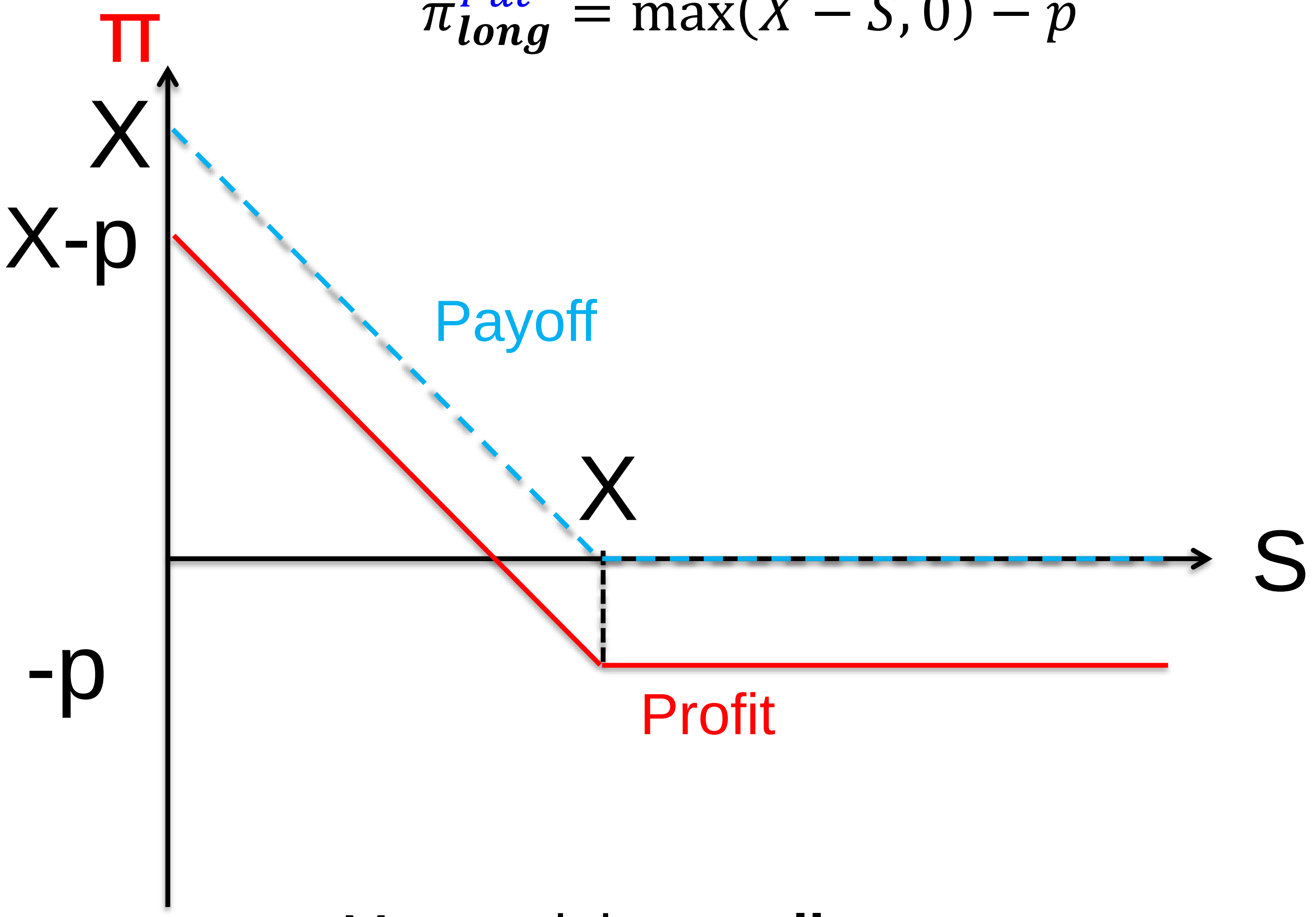
$$\pi_{long}^{Call} = \max(S - X, 0) - p$$



Memo: right to **buy**

Profit from *holding* (long) a Put

$$\pi_{long}^{Put} = \max(X - S, 0) - p$$



Memo: right to **sell**

Options: Profit for *writer* (short)

- ✓ **Call option**: writer is ***obliged*** *to* *sell* if the buyer of the call exercises the option
  - holder will exercise only if *S*>*X* , but he/she always pays the premium
  - hence writer's profit is:

$$\pi_{\boldsymbol{short}}^{Call} = p - max(S - X, 0)$$

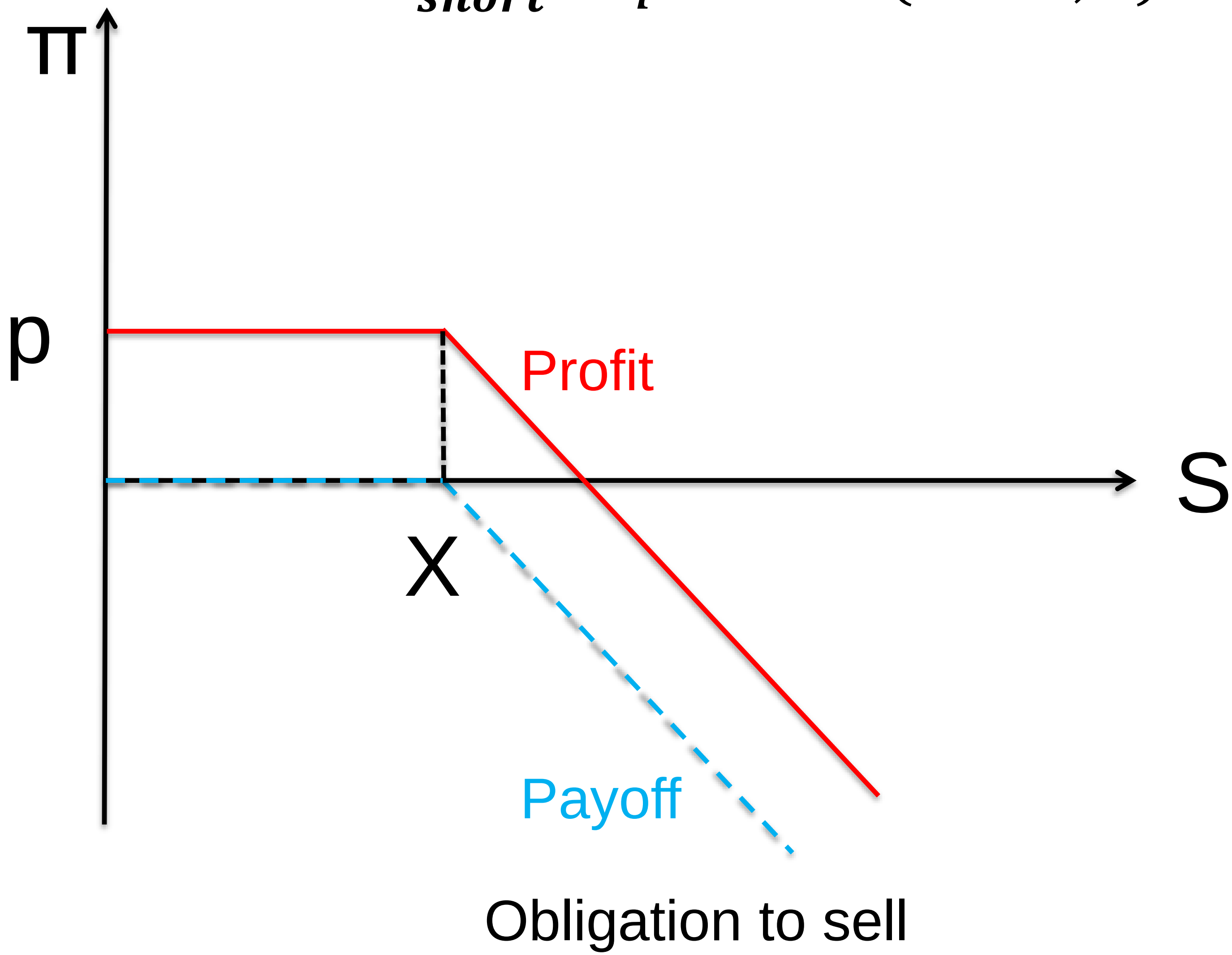
- ✓ **Put option**: writer is are *obliged* *to* *buy* if the buyer of the put exercises the option
  - holder will exercise only if *X*>*S*, but he/she always pays the premium
  - hence writer's profit is

$$\pi_{\boldsymbol{short}}^{Put} = p - max(X - S, 0)$$

Let's see everything graphically

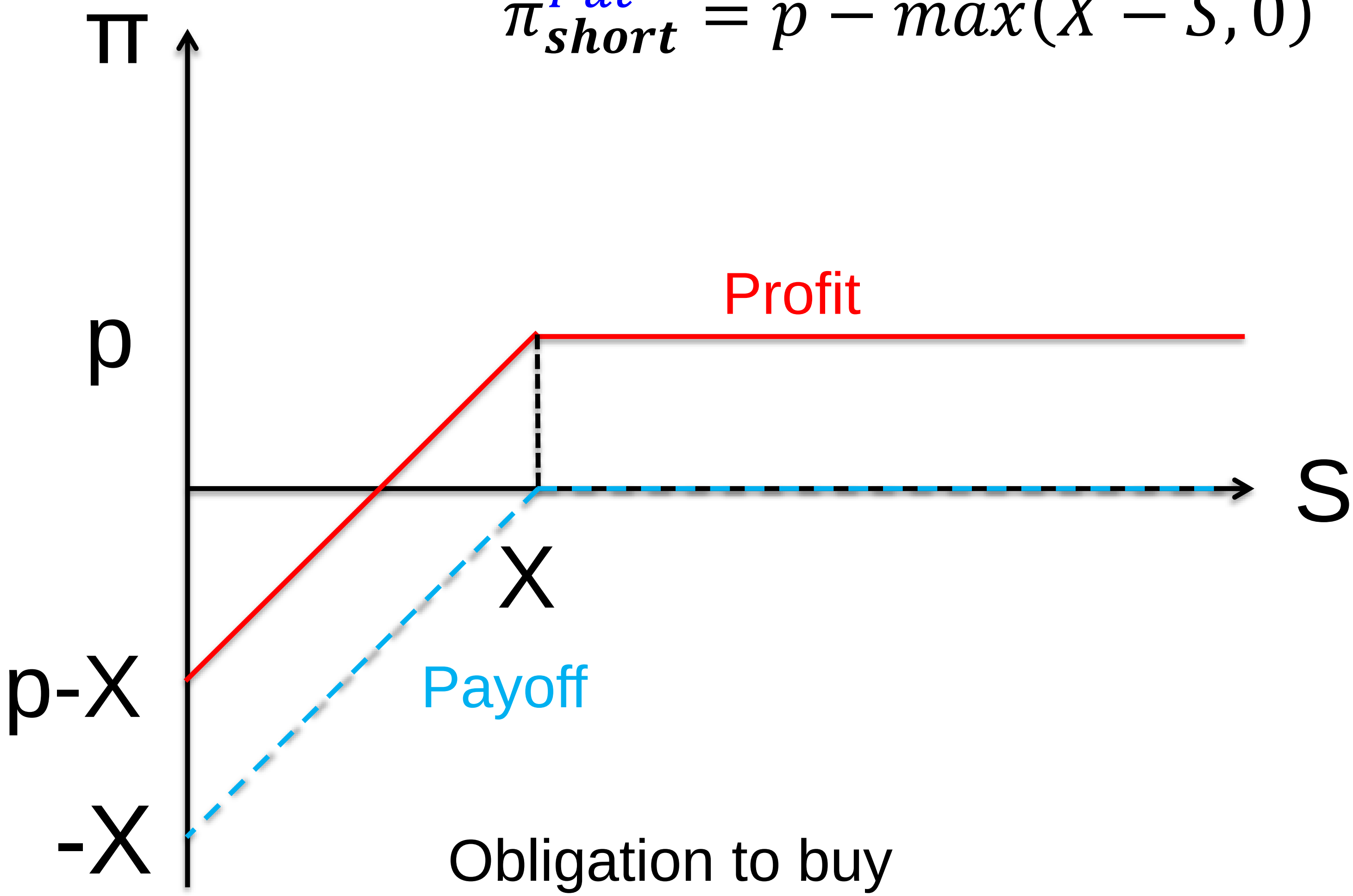
Profit from *writing* (short in) a Call

$$\pi_{short}^{Call} = p - max(S - X, 0)$$



Profit from *writing* (short in) a Put

$$\pi_{short}^{Put} = p - max(X - S, 0)$$



Obligation to buy

Examples of Options

- ✓ *Callable bonds*: bonds that can be called back by the issuing corporation (recall!)… they are a *call option*!
  - note that the corporation decides not to exercise the option if the price of the bond (spot) is below the face value (strike price)
  - the firm pays a premium in terms of higher *i*

- ✓ *Stock options:* a popular compensation scheme for managers
  - call options at a strike price and at specified dates
  - in theory … it should align the incentives of manager and of the company: pay linked to performance
  - sometimes not so efficient!

Options Styles

1. **European options:** do <u>not</u> allow early exercise (before expiration date)

2. **American options:** <u>allow</u> early exercise

   Note: the distinction between European and American options has nothing to do with where they are traded!

3. **Exotic options:** payoff structure is determined in a non-standard way
   - such as the average price of an underlying asset in a certain period of time (*Asian* option)
   - or *rainbow* options (on a basket of underlying assets) with payoffs determined by:
     - best of assets $\max(S_1, \dots, S_n, K)$
     - call on max $\max(\max(S_1, \dots, S_n,) - K, 0)$
     - …

Options Terminology

- ***In** the money*: exercise is profitable
- ***At** the money*: the strike equals the current price of the asset (the option's intrinsic value is zero),
- ***Out** of the money*: exercise is <u>un</u>profitable

- A call option is
  - ***in*** the money if S>X
  - ***at*** the money if S=X
  - ***out*** of the money if S<X

$$\pi_{long}^{Call} = \max(S - X, 0) - p$$

- A put option is
  - ***in*** the money if X>S
  - ***at*** the money if X=S
  - ***out*** of the money if X<S

$$\pi_{long}^{Put} = \max(X - S, 0) - p$$

perspective of the holder, not of the writer

Options Pricing (Premium)

- ✓ Pricing of options is a difficult business, need mathematical models (most famous *Black-Scholes-Merton*, economics Nobel prize winners in 1997)

- ✓ We only examine the determinants of the option premium:

  1. **Strike price** (X)
  2. **Spot price at expiration** (S_T)*
  3. **Volatility of underlying asset price**
  4. **Term to expiration**

*: specification of S "at expiration" only matters for European options, not for American ones.

Options Pricing Determinants

Increase in variable	Call option value (premium)	Put option value (premium)
Spot price (S_T)		
Strike price (X)		
Volatility		
Time to expiration		

Options Pricing Determinants

Increase in variable	Call option value (premium)	Put option value (premium)
Spot price (S_T)	↑	↓
Strike price (X)	↓	↑
Volatility	↑	↑
Time to expiration	↑	↑

Options Pricing Determinants

- ✓ The effects of increases in **strike and spot prices** (*S* and *X*) for call and put options have been examined in the previous slides

- ✓ Note the opposite effects for call and put

- ✓ *Both* **volatility** and **term to expiration** <u>increase</u> the value of both **call** and **put** options. Why?

- ✓ Look at the profit function from *holding* a call or a put option:

$$\pi_{long}^{call} = \max(S - X, 0) - p$$
$$\pi_{long}^{put} = \max(X - S, 0) - p$$

- ✓ There are <u>potentially large gains, but only limited losses</u> (-*p* at worst)

Options Pricing Determinants

- ✓ **Volatility**
  - can **only** have a **positive effect**:
  - the greater the chance of spot price (S) increase, the greater your expected profit
  - the greater chance of S decrease does not matter: you can always decide NOT to exercise the option

- ✓ **Term to expiration** (same logic)
  - it only increases the chance of a positive gain …
  - … with no impact on losses, capped at *-p*

- ✓ These two results are given by the *non-linearity* of the profit function (it is kinked!)
  - e.g.: can decide not to exercise option …not so for futures!

Options: Hedging Exercise

Let's see how hedging with options would work

- ✓ Suppose you have $100M in a stock portfolio indexed to the S&P500.

  Q: are you *long* or short in stocks? What type of risk do you want hedge?

Options: Hedging Exercise

Let's see how hedging with options would work

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  A: you're *long* in stocks; want hedge a fall in S&P500 price

Options: Hedging Exercise

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  Q: are you *long* or short in stocks? What type of risk do you want hedge?

  A: you're *long* in stocks; want hedge a fall in S&P500 price

- ✓ You can buy a put with strike price $90M or a call with strike price $110M. Both have a p=$1M

  Q: What do you buy to hedge?

Options: Hedging Exercise

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- ✓ Suppose you have $100M in a stock portfolio indexed to the S&P500.

  Q: are you *long* or short in stocks? What type of risk do you want hedge?

  A: you're *long* in stocks; want hedge a fall in S&P500 price

- ✓ You can buy a put with strike price $90M or a call with strike price $110M. Both have a p=$1M

  Q: What do you buy to hedge?

  A: you buy the put: guarantee that your portfolio is worth at least $90M ...

  … because you have the asset, so want to protect from downside risk (if you didn't have it, you'd buy a call)

Options: Example of NOT hedging

- ✓ In the previous example, had you bought the call, you'd be instead betting that the S&P500 will rise!
- ✓ This can increase your gains!

SoftBank's Bet on Tech Giants Fueled Powerful Market Rally

Japanese conglomerate led by billionaire Masayoshi Son placed billions in options bets on fast-rising tech stocks

- ✓ *SoftBank bought nearly $4 billion of shares in tech giants such as Amazon.com Inc., Microsoft Corp. and Netflix Inc. this spring, plus a stake in Tesla. Not included in those disclosures is the massive options trade […] SoftBank bought a roughly equal amount of call options tied to the underlying shares it bought, as well as on other names, according to people familiar with the matter.*
- ✓ *- WSJ, September 5th, 2020*

Options: Hedging Exercise

✓ Suppose the portfolio is worth $120M at maturity

Q: Do you exercise your put option? What is your profit?

Options: Hedging Exercise

✓ Suppose the portfolio is worth $120M at maturity

Q: Do you exercise your put option? What is your profit?

A: The put has strike price that is *lower* than spot at maturity ($X<S$ or \$100M<\$120M): choose <u>not</u> to exercise it

Profit from the option is –p =– $1M

Total profit (portfolio+option): (\$120M – \$100M) – \$1M= \$19M

Total value = \$120M - \$1M = \$119M

Had you **<u>not</u>** bought the option, total value would be $120

So, from \$100M to \$119M, not \$120M:
hedge value = \$120M - \$119M = \$1M

Options: Hedging Exercise

✓ Suppose the portfolio is worth $70M at maturity

Q: Do you exercise your put option? What is your profit?

Options: Hedging Exercise

✓ Suppose the portfolio is worth $70M at maturity

Q: Do you exercise your put option? What is your profit?

A: The put has strike price that is *higher* than spot at maturity (X>S or \$100M>\$70M): choose to exercise put

Profit from the option: X-S – p = \$90M - \$70M - \$1M = \$19M

Total profit (portfolio+option): (\$70 – \$100) + \$19 = -\$11M

Equivalently, since you sell for \$90: \$90 - \$100 - \$1= -\$11M

Total value = \$70M + \$19M = \$89M

Had you **not** bought the option, total value would be \$70

So from \$100M to \$89M, not \$70M:

hedge value = \$89M - \$70M = \$19M

Options: Hedging Exercise

Key idea of hedging with a put

- ✓ If things turn badly …
  - … you *capped* your portfolio value at $90M, by paying the premium of $1M
- ✓ If things turn well …
  - … you have also reduced your profits by <u>only</u> $1M.
- ✓ Put option here works like an *insurance*. This is why it is called ***protective put***

Options: Hedging Exercise

✓ Recap

	case (1) $S_T \geq X$	case (2) $S_T < X$
Spot price at t=0 (S_0)	100	100
Strike price (X)	90	90
Spot price at t=T (S_T)	120	70
Value of put option (π_{put})	-p= -1	X - S_T - p = 90 - 70 - 1 = 19
Total profit ($\pi_{TOT} = \Delta S + \pi_{put}$)	= (120 - 100) - 1 =19	=(70 - 100) + 19 = - 11
Final value ($S_0 + \pi_{TOT}$)	= 100 + 19 = 119	= 100 - 11 = 89
Final value (alternative)	(S_T - p) = 120 - 1 = 119	(X - p) = 90 - 1 = 89

Options: Hedging Exercise

- ✓ In general, the **additional value** of buying the **put** option compared to **not buying it** (where S_T is spot price of portfolio at maturity)

	$\boldsymbol{S_T < 90}$	$\boldsymbol{S_T \geq 90}$
Portfolio	S_T	S_T
Put	90-S_T-1	-1
Total	90-1	S_T-1

Example of speculation

Dortmund Bomb Suspect Attacked Soccer Team to Make $1 Million From Stock Drop

Suspect took out loan to finance the purchase of 15,000 ‘put options’ in popular soccer club Borussia Dortmund

11 April 2017

Recap on Options

✓ *Just check out the profit functions from before:*

$$\pi^{Call}_{\boldsymbol{long}} = \max(S - X, 0) - p$$

$$\pi^{Put}_{\boldsymbol{long}} = \max(X - S, 0) - p$$

potentially …
__limited__ *losses*
or large gains

holder

$$\pi^{Put}_{\boldsymbol{short}} = p - max(X - S, 0)$$

$$\pi^{Call}_{\boldsymbol{short}} = p - max(S - X, 0)$$

potentially …
__large__ *losses*
or limited gains

writer

Options Vs Futures

- ✓ Futures are **one-sided bets**:
  - you gain if the bet is correct and lose otherwise
  - same is true for the counterparty

- ✓ Options are **asymmetric bets**
- ✓ the **buyer** of the option will experience:
  - **large** gains if his/her bet is correct
  - **limited losses** if his/her bet is incorrect

- ✓ the **seller** of the option will experience:
  - **limited gains** if his/her bet is correct
  - **large losses** if his/her bet is incorrect

✓ *Just check out the profit functions from each case!*

Options Vs Futures

Once again:

- ✓ Future is an obligation for both parties
- ✓ Options
  - give a right to the option holder
  - but it is an obligation for the option writer (if the holder requests it)
- ✓ Futures are **costless** to initiate
  - no cash is exchanged until the position is closed
- ✓ Options are **not costless** to initiate
  - the buyer has to pay the seller an **option premium**